Linux IPTABLES cheatsheet

CHECKING IPTABLES RULES

```
# List all rules
$ iptables -L

# List rules with verbose output and numerical addresses
$ iptables -L -v -n

# List all rules in a format that can be reused
$ iptables -S
```

FLUSHING IPTABLES RULES

```
# List all rules
$ iptables -L

# List rules with verbose output and numerical addresses
$ iptables -L -v -n

# List all rules in a format that can be reused
$ iptables -S
```

SETTING DEFAULT POLICIES

```
# Set default policy for INPUT chain
$ iptables -P INPUT ACCEPT

# Set default policy for FORWARD chain
$ iptables -P FORWARD ACCEPT

# Set default policy for OUTPUT chain
$ iptables -P OUTPUT ACCEPT
```

WORKING WITH CHAINS

```
# Create a new chain
$ iptables -N <chain-name>

# Delete a user-defined chain
$ iptables -X <chain-name>

# Jump to a user-defined chain
$ iptables -A INPUT -j <chain-name>
```

ALLOWING/DENYING TRAFFIC WITH CONDITIONS

```
# Allow traffic from an IP on a specific port
$ iptables -A INPUT -p tcp -s <IP> --dport <port> -j ACCEPT

# Drop traffic from an IP on a specific port
$ iptables -A INPUT -p tcp -s <IP> --dport <port> -j DROP
```

DELETING RULES

```
# Delete a specific rule by its number
$ iptables -D INPUT <rule-number>

# Delete a specific rule
$ iptables -D INPUT -p tcp --dport <port> -j ACCEPT
```

ALLOWING/DENYING TRAFFIC

```
# Allow incoming traffic on a specific port
$ iptables -A INPUT -p tcp --dport <port> -j ACCEPT

# Allow traffic from a specific IP address
$ iptables -A INPUT -s <IP> -j ACCEPT

# Drop incoming traffic on a specific port
$ iptables -A INPUT -p tcp --dport <port> -j DROP

# Drop traffic from a specific IP address
$ iptables -A INPUT -s <IP> -j DROP

# Allow incoming traffic on a range of ports (e.g., 8000-8080)
$ iptables -A INPUT -p tcp --dport 8000:8080 -j ACCEPT
```

SAVING RULES, RESTORING RULES & LOGGING

```
# Save IPv4 rules to a file
$ iptables-save > /etc/iptables/rules.v4

# Restore IPv4 rules from a file
$ iptables-restore < /etc/iptables/rules.v4

# Log dropped packets
$ iptables -A INPUT -p tcp --dport <port> -j LOG --log-prefix "IPT-
ables-Dropped: " --log-level 4
```

NAT AND PORT FORWARDING

```
# Enable NAT on an interface
$ iptables -t nat -A POSTROUTING -o <interface> -j MASQUERADE

# Forward traffic from one port to another
$ iptables -t nat -A PREROUTING -p tcp --dport <port>
-j DNAT --to-destination <IP>:<port>

# Forward all HTTP (port 80) traffic to another local port (i.e, port 8080)
$ iptables -t nat -A PREROUTING -p tcp --dport 80 -j
REDIRECT --to-port 8080
```